

VPA IB DP CS — University Pathways Guide

□ VPA IB DP Computer Science — University Pathways Guide

Victoria Park Academy | IB DP Computer Science | University Guidance

A comprehensive guide to university pathways for VPA students completing IB DP Computer Science. Covers entry requirements, IB point requirements, course structures, and scholarship opportunities in the UK, US, China, and Hong Kong.

English: This guide helps VPA students and their families make informed decisions about university applications. It covers top institutions in four regions, with specific guidance for IB DP Computer Science students, including typical offers, required subjects, and preparation tips.

中文：本指南帮助VPA学生及其家庭做出明智的大学申请决定。涵盖四个地区的顶尖院校，为IB DP计算机科学学生提供具体指导，包括 typical offers、所需科目和准备建议。

Quick Reference: IB Points by Region

Region	Typical IB Score Required	HL Requirements	English Requirement
UK	38 – 42 points (top 10); 34 – 38 (Russell Group)	HL Maths + HL CS preferred; some require HL Physics	IELTS 6.5 – 7.5 or IB English A Lang/Lit Grade 5+
US	No fixed score; holistic review (typically 36+ for top 20)	Strong HL Maths and HL Science recommended	TOEFL 100+ or IELTS 7.0+ (if English not first language)
China	36 – 40 points (top universities); 32 – 36 (other 985/211)	HL Maths essential; HL Physics strongly preferred	HSK 5 – 6 for Chinese-taught programmes; IELTS 6.5+ for English-taught
Hong Kong	36 – 42 points (top 3); 32 – 36 (other universities)	HL Maths + HL CS preferred; HL Physics for Engineering	IELTS 6.5 – 7.0 or IB English A Lang/Lit Grade 5+

United Kingdom UK

University of Cambridge

Cambridge, England | Founded 1209

Course: BA Computer Science (3 years) or MEng Computer Science (4 years)

Ranking: QS World #2 Computer Science (2024)

Entry Requirements:

IB: 40 – 42 points with 776 at HL

HL Mathematics: Analysis and Approaches (AA) required at Grade 7

HL Physics or HL Computer Science strongly preferred

STEP (Sixth Term Examination Paper) may be required for some colleges

Admissions assessment: TMUA (Test of Mathematics for University Admission) or CSAT

Scholarships: Cambridge International Scholarships (full fees + living); College-specific bursaries; Gates Cambridge (postgraduate only)

VPA Prep: Start TMUA preparation in G11; aim for HL Maths AA Grade 7; participate in UKMT/BMO competitions; read "The New Turing Omnibus" by A.K. Dewdney

University of Oxford

Oxford, England | Founded 1096

Course: BA Computer Science (3 years) or MCompSci (4 years)

Ranking: QS World #5 Computer Science (2024)

Entry Requirements:

IB: 39–42 points with 766 at HL

HL Mathematics: AA required at Grade 7

HL Further Mathematics or HL Physics preferred

MAT (Mathematics Admissions Test) required — taken in October of G12

Interview required (typically December)

Scholarships: Oxford Pershing Square Scholarship; Reach Oxford Scholarship (developing countries); College scholarships

VPA Prep: MAT preparation from G11 summer; practice MAT past papers under timed conditions; prepare for Oxford-style tutorial interviews; read "Algorithmic Thinking" by Daniel Zingaro

Imperial College London

London, England | Founded 1907

Course: BEng Computing (3 years) or MEng Computing (4 years)

Ranking: QS World #7 Computer Science (2024)

Entry Requirements:

IB: 39–42 points with 7 in HL Mathematics (AA)

HL Physics or HL Computer Science at Grade 6+

Most offers require A*A*A equivalent (IB 40+ with 776 HL)

Admissions test: TMUA or STEP may be required

Scholarships: Imperial College London Scholarships (merit-based, up to £5,000/year); President's Undergraduate Scholarships (full fees); Royal Commission for the Exhibition of 1851

VPA Prep: Strong focus on HL Maths and HL Physics; complete coding projects beyond the curriculum; consider Imperial's Global Summer School; apply for the RISE programme

University College London (UCL)

London, England | Founded 1826

Course: BSc Computer Science (3 years) or MEng Computer Science (4 years)

Ranking: QS World #22 Computer Science (2024)

Entry Requirements:

IB: 38–40 points with 7 in HL Mathematics (AA)

HL Computer Science or HL Physics at Grade 6+

English requirement: IELTS 7.0 (7.0 in writing) or IB English A Grade 5+

No admissions test required

Scholarships: UCL Global Undergraduate Scholarship (full fees); UCL Hong Kong Scholarship; Access Opportunity Scholarship

VPA Prep: UCL values interdisciplinary study — consider combining CS with another subject (e.g., CS + Philosophy, CS + Business); strong IA project is advantageous

United States US

Massachusetts Institute of Technology (MIT)

Cambridge, Massachusetts | Founded 1861

Course: BS Computer Science and Engineering (4 years)

Ranking: QS World #1 Computer Science (2024)

Entry Requirements:

Holistic review — no minimum IB score, but successful applicants typically have 40+

HL Mathematics (AA) and HL Physics strongly recommended

SAT/ACT not required for 2024–2025 cycle (test-optional)

Essays (5 supplemental MIT essays) are critical — demonstrate passion for CS

Letters of recommendation (2 teachers + 1 counsellor)

Interview with MIT alumni (strongly recommended)

Financial Aid: Need-blind for all applicants; meets full demonstrated need; no loans — all aid is grants/scholarships. Average family contribution for families earning under \$75,000: \$0

VPA Prep: MIT values hands-on makers — build projects, contribute to open source, participate in hackathons; take MIT OpenCourseWare classes; apply to MITES (Minority Introduction to Engineering and Science) summer programme; prepare for the "What do you do for fun?" essay

Stanford University

Stanford, California | Founded 1885

Course: BS Computer Science (4 years) — also offers CS + X joint majors

Ranking: QS World #3 Computer Science (2024)

Entry Requirements:

Holistic review; successful applicants typically IB 40+

HL Mathematics (AA) and HL Science recommended

SAT/ACT test-optional

8 short essays (the "Stanford supplement") — intellectual vitality, community contribution, meaningful extracurriculars

Teacher recommendations (2)

Optional arts/tech portfolio

Financial Aid: Need-blind for US citizens; need-aware for international students; meets full demonstrated need. Families under \$100,000: no parent contribution expected

VPA Prep: Stanford values "intellectual vitality" — show curiosity beyond the curriculum; start a CS-related initiative at VPA; participate in Stanford Pre-Collegiate Summer Institutes; read CS papers and blog about them

Carnegie Mellon University (CMU)

Pittsburgh, Pennsylvania | Founded 1900

Course: BS Computer Science (4 years) — also offers interdisciplinary programmes (CS + Arts, CS + Business)

Ranking: QS World #4 Computer Science (2024)

Entry Requirements:

IB: 38–42 points recommended

HL Mathematics (AA) at Grade 6+; HL Computer Science or HL Physics preferred

SAT/ACT test-optional but recommended for international students

Supplemental essays (3) + Common App essay

Portfolio of programming projects (optional but strongly recommended)

Financial Aid: Merit-based scholarships available (Presidential Scholarships, full tuition); need-based aid available

VPA Prep: CMU is known for practical, project-based CS — build a strong GitHub portfolio; participate in competitive programming (USACO, Codeforces); CMU's CS Academy offers free Python courses; consider the CMU Pre-College Programme

China China

Tsinghua University 清华大学

Beijing, China | Founded 1911

Course: 计算机科学与技术 (Computer Science and Technology) — 4 years

Ranking: QS World #11 Computer Science (2024); #1 in Asia

Entry Requirements (International Students):

IB: 38–42 points

HL Mathematics (AA) at Grade 7; HL Physics at Grade 6+

Chinese language: HSK 5 (for Chinese-taught programmes) or IELTS 6.5+ (for English-taught programmes)

Tsinghua International Student Entrance Exam (some programmes)

Interview (in Chinese or English depending on programme)

Scholarships: Chinese Government Scholarship (full coverage); Beijing Government Scholarship; Tsinghua University Scholarship (partial to full); Belt and Road Scholarship

VPA Prep: Strong HSK preparation essential for Chinese-taught programmes; participate in Olympiad informatics competitions; Tsinghua values mathematical rigour — aim for HL Maths Grade 7; consider Tsinghua's summer camp for international students

Peking University 北京大学

Beijing, China | Founded 1898

Course: 计算机科学与技术 (Computer Science) — 4 years

Ranking: QS World #15 Computer Science (2024)

Entry Requirements (International Students):

IB: 38–42 points

HL Mathematics (AA) at Grade 7; HL Physics preferred

HSK 5 for Chinese-taught; IELTS 6.5+ for English-taught

PKU International Student Entrance Exam (数学 and 物理)

Interview

Scholarships: Chinese Government Scholarship; PKU International Student Scholarship; Yenching Academy Scholarship (for postgraduate)

VPA Prep: PKU's Yuanpei College allows flexible major selection — consider applying through Yuanpei if unsure about CS specifically; strong mathematical foundation is essential; prepare for the entrance exam in Chinese

Shanghai Jiao Tong University (SJTU) 上海交通大学

Shanghai, China | Founded 1896

Course: 计算机科学与技术 (Computer Science) — 4 years

Ranking: QS World #27 Computer Science (2024)

Entry Requirements (International Students):

IB: 36–40 points

HL Mathematics (AA) at Grade 6+; HL Physics or HL Computer Science preferred

English-taught programmes available (IELTS 6.5+ or TOEFL 90+)

SJTU entrance exam or interview may be required

Scholarships: Shanghai Government Scholarship (Class A: full; Class B: partial); SJTU International Student Scholarship; Chinese Government Scholarship

VPA Prep: SJTU's Zhiyuan College offers elite CS programme with English instruction; strong industry connections in Shanghai's tech sector; consider SJTU's summer research programme for international students

Hong Kong Hong Kong

University of Hong Kong (HKU) 香港大学

Hong Kong | Founded 1911

Course: BEng Computer Science (4 years) or BSc Computer Science (4 years)

Ranking: QS World #43 Computer Science (2024)

Entry Requirements:

IB: 36–40 points

HL Mathematics (AA) at Grade 5+; HL Physics or HL Computer Science preferred

IELTS 6.5+ or IB English A Lang/Lit Grade 5+

Interview may be required for some applicants

Scholarships: HKU Foundation Scholarship (full tuition + living); HKU International Student Scholarship; Faculty of Engineering Scholarships

VPA Prep: HKU values well-rounded students — balance academic excellence with extracurriculars; apply for HKU's Summer Institute; consider the HKU-SCF FinTech Academy for finance+CS interest

Hong Kong University of Science and Technology (HKUST) 香港科技大学

Hong Kong | Founded 1991

Course: BEng Computer Science (4 years) or BSc Computer Science (4 years)

Ranking: QS World #40 Computer Science (2024)

Entry Requirements:

IB: 36–40 points

HL Mathematics (AA) at Grade 5+; HL Physics or HL Computer Science preferred

IELTS 6.0+ or IB English A Lang/Lit Grade 5+

Interview for shortlisted candidates

Scholarships: HKUST International Scholarship (full tuition + living); HKUST Entrance Scholarship (merit-based, partial to full); School of Engineering Scholarships

VPA Prep: HKUST has strong industry ties with Silicon Valley and Shenzhen tech companies; apply for HKUST's Summer Research Programme; strong programming portfolio is advantageous; consider the Dual Degree Programme with UC Berkeley or Yale

Application Timeline for IB Students

Date	UK (UCAS)	US (Common App)	China	Hong Kong
June – Aug (G11)	Research courses; attend open days (virtual/physical)	Begin Common App; draft personal statement	Research programmes; check HSK requirements	Research programmes; attend information sessions
Sept – Oct (G12)	UCAS application opens (1 Sept); submit by 15 Oct (Oxbridge/Medicine) or 15 Jan	Early Decision/Action deadlines (1 Nov, 15 Nov)	Application opens; prepare documents	Application opens (Sept); early round deadlines
Nov – Dec (G12)	Admissions tests (TMUA, MAT, STEP); interview invitations	Regular Decision deadlines (1 Jan – 15 Jan); interviews	Submit application; entrance exam registration	Submit application; interview invitations
Jan – Mar (G12)	Interviews (Oxbridge: Dec; others: Jan – Mar); offers begin	Financial aid applications; alumni interviews	Entrance exams; interviews	Interviews; offers begin
Apr – May (G12)	Final offers; Firm and Insurance choices by June	Decision letters (Ivy Day: late March; RD: April)	Admission results; visa application	Final offers; accept by deadline
July (G12)	Results day; Clearing if needed	—	—	—

Subject Combination Advice

Ideal IB DP Subject Combination for CS:

HL: Mathematics: Analysis and Approaches (essential), Computer Science (strongly recommended), Physics or Chemistry (recommended for some programmes)

SL: English A Language and Literature, a second language, a humanities subject (Economics, Psychology, or Philosophy)

EE: Computer Science Extended Essay is highly regarded by admissions tutors

Alternative for students not taking HL CS: HL Maths AA + HL Physics + one other HL science is acceptable for most programmes, but HL CS is advantageous for CS-specific courses.

Scholarship Application Tips

Start early: Many scholarship deadlines are earlier than admission deadlines (e.g., Gates Cambridge: October; Rhodes: October)

Track record matters: Scholarship committees look for sustained impact, not just grades — show leadership, community service, and initiative

Essays are critical: Most scholarships require additional essays — start drafting in G11 summer

Recommendations: Ask teachers who know you well and can speak to your character, not just academic ability

Financial documentation: Need-based scholarships require detailed family financial information — prepare early

奖学金申请建议：尽早开始——许多奖学金截止日期早于录取截止日期；成绩记录很重要——奖学金委员会看重持续影响力，而不仅仅是成绩；论文至关重要——大多数奖学金需要额外论文，G11暑假开始起草；推荐信——请了解你并能评价你品格的老师撰写；财务文件——需要奖学金要求详细的家庭财务信息——提前准备。

Gap Year & Deferral Options

UK: Most universities allow deferral for 1 year; gap year activities should be meaningful (internships, research, travel, volunteering)

US: Deferral policies vary; some universities encourage gap years (Harvard, MIT, Yale)

China: Deferral is generally not permitted; students must apply in the year they intend to enrol

Hong Kong: Most universities allow 1-year deferral with valid reason

☐ Victoria Park Academy | IB DP Computer Science | University Pathways Guide v1.0 Information accurate as of 2024.
Always verify current requirements with university admissions offices.